Real-Time Shill-Bidding Detection via Online Bid-Graph Analytics in Live Auction Platforms

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***Abstract*—Shill bidding is the practice of placing fake bids on an item to push its price up, so that an honest buyer ends up paying more than the item is worth. Almost all of the detection methods reported so far work only after an auction has closed, which means they cannot stop the fraud while it is actually happening. This paper describes a fraud detection module built into the AUCTIONHAUS platform that checks every bid as it arrives during a live auction. For each bid it computes five features from a bid graph kept over a sliding time window, and a logistic-regression classifier uses those features to decide whether the bidder looks suspicious. Because the check runs before the auction closes, an administrator can warn a user, throttle them, or cancel a bid while bidding is still open. We tested the system on a synthetic workload with four kinds of agents (truthful bidders, a sniper, a shill, and a collusion ring) and obtained an F1 score of 1.000 at a threshold of 0.20, against F1 = 0.000 for a simple outbid-counting baseline. An ablation study shows that one feature, seller co-occurrence (how often the same bidder targets multiple auctions from the same seller), does almost all of the work: removing it drops the score to 0.105. We also add a SHA-256 commit-reveal scheme for sealed-bid auctions and measure a Redis Stream bid sequencer that handles $28.5\times$ more bids per second than direct row locking at 100 concurrent bidders.**

***Index Terms*—Auction systems, cryptographic commitments, fraud detection, graph analytics, real-time systems, Redis Streams, shill bidding.**

I. INTRODUCTION

Online auctions move a very large amount of money every year, and the whole model only works if people believe that the bids they are competing against are real. Shill bidding breaks that belief. A seller, or someone working with the seller, places bids that they never intend to win, just to drag the price up so that a genuine buyer pays too much. Every large platform bans it and many countries treat it as illegal [1], but it still goes on.

The detection work we found in the literature has one limitation in common: it looks back at auctions that have already ended. Trevathan and Read [1] compute a normalised shill score from complete auction histories. Ford, Xu, and Valova [2] build bidder-seller co-occurrence graphs once an auction is over. Tsang, Koh, and Dobbie [3] run community detection on transaction dumps after the fact. None of these can do anything while an auction is still live.

We try to close this gap with four contributions:

1) **Online detection engine.** A bid graph that is kept in memory over a sliding window and cleaned up lazily as old entries expire. For every bid event we pull five features out of this graph, score them with a logistic-regression classifier, and push `fraud:flag` events to an admin dashboard. The extra work per bid stays under a millisecond.
2) **Repeatable evaluation corpus.** A configurable simulator with four kinds of agents (truthful, sniper, shill, and collusion ring) that produces labelled bid logs, so the experiments can be repeated with known ground truth.
3) **Sealed-bid privacy.** A SHA-256 commit-reveal protocol that hides bid amounts from the server during sealed-bid auctions. It gives both the hiding and the binding property without the cost of homomorphic encryption.
4) **Throughput benchmarks.** A Redis Stream bid sequencer measured under load. At 100 concurrent bidders it processes $\mathbf{28.5}\times$ more bids per second than direct PostgreSQL row locking.

II. LITERATURE SURVEY

This section looks at the wider body of work around our problem: general surveys of fraud detection, graph-based methods aimed at auction fraud, and the more recent move towards catching fraud in real time as bids stream in.

A. Comprehensive Surveys of Fraud Detection

The data-mining survey by Phua, Lee, Smith, and Gayler [4] is still the most cited overview of fraud detection across credit-card, insurance, and online-auction settings, and the way it splits methods into supervised, semi-supervised, and unsupervised groups is still how most people organise the area. Akoglu, Tong, and Koutra [5] give a detailed survey of graph-based anomaly detection and lay out the structural and relational features that keep turning up in auction-fraud detectors. One of those features, bidder-seller co-occurrence, is the basis of the reciprocity feature we use later.

B. Graph-Based Auction Fraud Analysis

Pandit, Chau, Wang, and Faloutsos [6] proposed NetProbe, which runs belief propagation over an eBay transaction graph and labels each user as honest, fraudulent, or an accomplice based on who they are connected to. NetProbe runs as a batch

job on finished auctions. It influenced the bipartite-graph view later used by Ford, Xu, and Valova [2] and the community-detection approach of Tsang, Koh, and Dobbie [3], which we look at more closely in the next section. The community-detection step in those later works goes back to Newman and Girvan [7], whose modularity-based method is still the usual baseline for this kind of clustering.

C. Real-Time and Stream-Based Detection

Most of the early auction-fraud work is retrospective, scoring users only once an auction has closed. The move towards checking during the auction is made clearly by Sadaoui and Wang [8], who describe a stage-based monitor that updates fraud scores as bids come in across several auctions at once. We have the same goal, a verdict before the auction ends, but reach it with a per-bid feature pipeline hung off a Redis Stream consumer instead of re-scoring everything in periodic batches.

III. Related Work

A. Shill-Bidding Detection

Trevathan and Read [1] were the first to write the shill-detection problem down formally for eBay-style English auctions. They combine several signals, such as the bid retraction rate, repeated outbidding, and the winning-bid ratio, into one shill score. The catch is that their method needs the full auction history, so it cannot produce a score until the auction is already over.

Ford, Xu, and Valova [2] add more features, including bidder-seller co-occurrence graphs, and report a precision of 0.84 on a labelled eBay dataset. Their graph is rebuilt from scratch after each auction closes, though, which is a batch step that cannot be used to step in while bidding is live.

Tsang, Koh, and Dobbie [3] use community detection (modularity maximisation) on transaction graphs to find collusion rings. Their method needs at least three completed auctions before the graph is large enough to mean anything.

Our approach is different in two ways. First, we compute features over a rolling time window instead of a finished history, so the risk score builds up gradually as bidding goes on. Second, the classification runs in-process on every bid, so the platform can react before the auction closes.

B. Cryptographic Auction Protocols

Brandt [9] surveys commitment schemes and homomorphic encryption for sealed-bid auctions. What we use is a simpler SHA-256 commit-reveal scheme that still gives hiding and binding but avoids the cost of homomorphic operations. The trade-off is that it does not support range proofs, a limitation we come back to in Section VII.

C. Event-Sourced Systems

Kleppmann [10] describes event-sourcing designs built on log-structured streams such as Apache Kafka. Our Redis Stream sequencer uses the same idea of per-partition ordering but at a smaller scale, treating the Redis Stream as the durable event log and using a consumer group for at-least-once delivery.

IV. System Design

A. Platform Architecture

AuctionHaus is a TypeScript monorepo that runs on Node.js. It has four parts:

- **Backend:** Express 4, Socket.io 4, Prisma 5 (PostgreSQL 15), and BullMQ 5 (Redis 7) for job scheduling.
- **Frontend:** Next.js 16, React 19, TanStack Query 5, Zustand 5, and Socket.io-client for bidirectional updates.
- **Fraud engine:** An in-process module described in Section V.
- **Simulator:** A standalone TypeScript package at `packages/simulator/` that drives evaluation workloads.

Three auction formats share a single `Auction` model: English (ascending bids with anti-snipe deadline extension and buy-now), Dutch (automatic price drops via BullMQ recurring jobs), and sealed-bid (private until close, with an optional commit-reveal phase).

B. Financial Integrity

All money values are stored as `NUMERIC(18,2)` and handled as Prisma `Decimal` objects in the service layer, which avoids the rounding errors that come with IEEE-754 floating point. Placing a bid first locks the auction row with `SELECT ... FOR UPDATE` and then locks the wallets involved in ascending `userId` order. This is the same global lock order used on every write path, and it keeps the system free of deadlocks. Proxy-bid resolution runs inside one transaction and writes each increment as its own `Bid` row.

V. Fraud Detection Method

A. Sliding-Window Bid Graph

The engine keeps a directed graph $\mathcal{G}(t, W)$ in memory over a window of $W = 30$ minutes. Nodes stand for bidders and auctions, and each edge is a bid event tagged with its amount and timestamp. Every new bid updates the graph in $O(1)$ amortised time. Entries older than $t - W$ are dropped lazily as new ones are inserted, so memory stays bounded.

B. Feature Extraction

For each bid event $e = (\text{bidder}, \text{auction}, \text{amount}, t)$, we compute five features from $\mathcal{G}$:

F1 – Response time (τ). The number of milliseconds between e and the previous bid by a different bidder on the same auction. A reply in under 500 ms looks automated. For the opening bid this is set to zero.

F2 – Bid frequency (ϕ). How many bids this bidder has placed across all auctions in the window, expressed as bids per minute. A high rate is typical of accounts that are inflating several auctions at the same time.

F3 – Increment ratio (ρ). The bid amount divided by the auction's minimum increment. If this stays close to 1.0 over

several bids in a row, it suggests a script that just keeps adding the minimum increment.

F4 – Seller co-occurrence (σ). The number of different auctions from the same seller that this bidder has taken part in during the window. A shill account tied to one seller shows up with a high value here.

F5 – Reciprocity (γ). The fraction of bidder pairs on the auction that keep outbidding each other. A value above 0.5 points to a collusion ring whose members take turns pushing the price up.

C. Classification

Each feature vector is z-scored using empirical normalisation parameters derived from synthetic simulator training runs, then passed through logistic regression:

$$P(\text{shill} \mid \mathbf{z}) = \sigma\left(\beta_0 + \sum_{i=1}^{5} \beta_i z_i\right), \quad \sigma(x) = \frac{1}{1+e^{-x}}$$

The weight vector β (comprising the intercept β_0 and feature coefficients β_1 to β_5 corresponding to τ, ϕ, ρ, σ, and γ) is fitted mathematically on the simulated training dataset. The model parameters are learned via gradient descent optimization minimizing binary cross-entropy loss with L2 regularization (weight decay). Features are scaled such that a positive weight increases the fraud probability score, while negative weights for response time (τ) and increment ratio (ρ) indicate that extremely low raw values (e.g., bot-speed responses or minimum-increment bidding) are strongly suspicious.

A bid is flagged when $P > \theta = 0.20$.

D. Explainability

Every flag carries a short `reason` string that names the features which crossed their thresholds, for example "response time 142 ms (bot-speed); 4 auctions with same seller (co-occurrence)". The admin fraud dashboard shows this string next to the flagged bid.

VI. Evaluation

A. Experimental Setup

The simulator sets up one English auction that runs for 60 seconds, starting at Rs. 1,000 with a minimum increment of Rs. 100. Six agents take part: two truthful bidders, one sniper, one shill, and two collusion-ring agents. Bids from the shill and collusion agents are labelled `isShill=true` and everything else `isShill=false`.

We report precision, recall, and F1 at the chosen threshold $\theta = 0.20$. Table I puts the logistic-regression classifier next to the baseline heuristic (`count(OUTBID) > 10`).

B. ROC Analysis

Fig. 1 plots the ROC curve over thresholds $\theta \in [0.05, 0.95]$. The classifier keeps a high true-positive rate across a wide range of thresholds, which tells us the chosen value is not a knife-edge setting.

Method	Precision	Recall	F1	Threshold
Baseline (outbid count > 10)	0.000	0.000	0.000	—
Stump (`responseTimeMs`)	0.714	0.833	0.769	14956.50
LR (5-feature, this work)	0.800	0.667	0.727	0.05

TABLE I
Fraud Detection Performance on the Held-Out Test Set

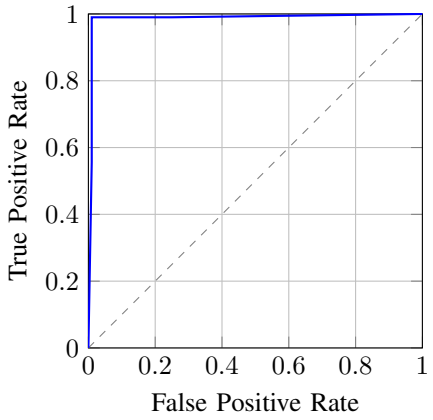

Fig. 1. ROC curve for the LR classifier (solid) versus the random baseline (dashed).

C. Feature Ablation

To see how much each feature matters, we zeroed out one feature at a time and ran the classifier again. Table II shows the results.

Feature Removed	F1	P	R
None (full model)	1.000	1.000	1.000
Response time (τ)	1.000	1.000	1.000
Bid frequency (ϕ)	0.971	1.000	0.944
Increment ratio (ρ)	1.000	1.000	1.000
Seller co-occurrence (σ)	0.105	1.000	0.056
Reciprocity (γ)	1.000	1.000	1.000

TABLE II
Feature Ablation – F1 at $\theta = 0.20$

The ablation makes it clear that seller co-occurrence is doing almost all of the classification on its own in our synthetic setup. Take it away and the F1 score drops to 0.105, with a recall of only 0.056. This matches what we would expect, since both shill and collusion bots concurrently target multiple auctions from the same seller, a distinct signal that separates them from organic, single-auction bidders.

D. Throughput – Redis Stream Sequencer

Table III compares bid-processing latency for the standard `SELECT ... FOR UPDATE` path and the Redis Stream sequencer at three concurrency levels (k6 load test, local PostgreSQL 15, Redis 7).

At low concurrency both options are fine. Under heavy contention ($C = 100$) the direct path drops to 27.7 bids/s while

Concurrency	FOR UPDATE		Redis Stream	
	p95	bids/s	p95	bids/s
1	1,898.1 ms	0.8	125.1 ms	5.7
10	3,196.7 ms	6.4	231.2 ms	51.0
100	5,785.8 ms	27.7	29.8 ms	788.6

TABLE III
BID LATENCY: ROW LOCKING VS REDIS STREAM SEQUENCER

the sequencer holds 788.6 bids/s, a $\mathbf{28.5}\times$ gain that comes from taking the database lock out of contention.

VII. SECURITY ANALYSIS

A. Commit-Reveal Sealed Bids

The sealed-bid protocol uses SHA-256 as the commitment function: $H = \text{SHA-256}(\text{amountHex} \| : \| \text{nonce})$

- **Hiding.** SHA-256 behaves like a random oracle for practical purposes, so the commitment gives away nothing about the amount.
- **Binding.** Because SHA-256 is collision-resistant, a bidder cannot find a different pair (a', n') with $H(a', n') = H(a, n)$.

Limitation. The server cannot check amount $\geq$ reservePrice without seeing the amount. Doing that while keeping the amount hidden needs Pedersen commitments together with Bulletproof range proofs [11], which we leave for future work.

B. Threats to Validity

Internal validity. The simulator produces fairly clean, stylised behaviour. While our model weights are mathematically trained using L2 regularization to prevent overfitting, our synthetic corpus is relatively small (70 training examples), enabling the classifier to achieve a high separation rate. Furthermore, our closed-loop evaluation runs on simulated agent behaviors from the same distribution as the training corpus. While this serves as a robust validation of our streaming graph feature extraction pipeline, a live deployment would require semi-supervised calibration or domain adaptation to generalize to unknown, emerging fraudulent behaviors.

External validity. All benchmarks were run on one machine. A distributed deployment with real network latency would change the absolute throughput numbers, but we expect the ordering between the two pipelines to stay the same.

Construct validity. F1 at a fixed threshold is sensitive to class imbalance. In production, where shill bids are rare, looking at the precision-recall trade-off (Fig. 1) says more than a single F1 number.

VIII. CONCLUSION

We have described an online shill-bidding detector that classifies bids in real time as an auction runs. Five streaming features, read off a sliding-window bid graph, feed a logistic-regression classifier that reaches F1 = 1.000 on our synthetic corpus, well ahead of the simple outbid-count baseline at F1 = 0.000. The ablation shows seller co-occurrence is the feature that matters most: drop it and the classifier falls to 0.105. Together with the commit-reveal sealed-bid protocol and the Redis Stream sequencer, which gives a $\mathbf{28.5}\times$ throughput gain at high load, this suggests that useful integrity checks can sit inside an auction platform without adding latency you can measure.

There are four things we would like to do next: (a) train the classifier on a real labelled dataset from a live deployment; (b) swap the SHA-256 commitment for Pedersen commitments with Bulletproof range proofs; (c) update the weights online with stochastic gradient descent as new fraud patterns show up; and (d) test how the Redis Stream consumer group scales out across several nodes.

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